***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 24 Microbial Symbioses with Humans

24.1 Multiple Choice Questions

1) What is TRUE about stomach microbiota?
A) Few microbes can survive because it is so acidic.
B) The alkaline environment supports a wide variety of species.
C) Despite the acidic of the stomach, diverse phylotypes of resident microbes thrive in the stomach.
D) There is a rich diversity of transient microbes, but few resident microbes.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

2) The pouch at the beginning of the large intestine is called the
A) cardia.
B) ileum.
C) colon.
D) cecum.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.2

3) One microenvironment of the skin is an area where glands produce an oily substance called
A) mucus.
B) sebum.
C) fimbrae.
D) lipid A.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.5

4) Following antibiotic therapy, patients are often administered ________ to facilitate recolonizaton of normal microbiota in the intestines.
A) fluoride
B) iron
C) probiotics
D) antivirals
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.11

5) The damaged areas of teeth caused by organic acids produced by dental plaque are called
A) degenerative plaque.
B) calcified plaque.
C) periodontal disease.
D) dental caries.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

6) Which of the following is NOT true about breast fed infants compared with those fed formula?
A) They have greater numbers of *Bifidobacterium longum.*
B) They have greater numbers of *Clostridium difficile.*
C) They have greater numbers of *Citrobacter spp.*
D) They have greater numbers of *Enterobacter cloacae.*
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.7

7) Which of these microorganisms is most likely to be found in the human stomach?
A) *Helicobacter pylori*
B) *Streptococcus sobrinus*
C) *Streptococcus mutans*
D) *Roseobacter denitrificans*
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.2

8) Normal microbiota in the duodenum is
A) similar to the microbiota in the stomach.
B) dominated by aerobic organisms.
C) tolerant to highly alkaline environments.
D) tolerant to high salinity.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

9) Which of the following are NOT found in the gastrointestinal tract of healthy humans?
A) *Bacteroides*
B) *Clostridium*
C) *Escherichia coli*
D) protists
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

10) The following compounds are all produced by intestinal microbiota EXCEPT
A) thymine.
B) vitamin C.
C) vitamin B_{12}.
D) vitamin K.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

11) Vaginitis commonly results from overgrowth of
A) *Candida.*
B) *Trichomonas vaginalis.*
C) *Candida* and *Trichomonas vaginalis.*
D) Neither *Candida* nor *Trichomonas vaginalis.*
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

12) Only particles smaller than ________ μm in diameter reach the lungs.
A) 10
B) 20
C) 30
D) 40
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.3

13) The vagina of an adult female is
A) highly acidic.
B) highly alkaline.
C) weakly acidic.
D) weakly alkaline.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.4

14) What is the difference between vaginitis and vaginosis?
A) Vaginitis is more serious and vaginosis is subclinical.
B) Vaginosis is more serious and vaginitis is subclinical.
C) Vaginosis occurs first and then develops into vaginitis.
D) Vaginosis is an acute infection and vaginitis is always a chronic infection.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.9

15) Children with severe dental caries tend to have elevated levels of all of the following bacteria EXCEPT
A) *Aestuariimicrobium*.
B) *Streptococcus*.
C) *Granulicatella*.
D) *Actinomyces*.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

16) Individuals who have chronic periodontitis tend to have
A) lower microbial diversity than those without.
B) significantly greater microbial diversity than those without.
C) slightly greater microbial diversity than those without.
D) unpredictable levels of microbial diversity in comparison to those without chronic periodontitis.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

17) What is the role of fecal transplants in medicine?
A) They can provide a healthy microbiota to help treat *Clostridium difficle* infections.
B) They act as prebiotics to encourage healthy microbial growth.
C) They remove the unhealthy microbes, allow the individual an opportunity to have recolonization by non-pathogens ingested with food.
D) They can be used to treat infections caused by the overgrowth of *Lactobacilli*.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.7, 24.11

18) Common probiotics include
A) *Bifidobacterium* species.
B) *Clostridium* species.
C) *Escherichia coli*.
D) *Corynebacterium* species.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.11

19) A microbial community that does not change when exposed to stressors is called
A) resilient.
B) resistant.
C) robust.
D) regimented.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

20) *Clostridium difficile* infections most often occur
A) following antibiotic treatment.
B) following a fecal transplant.
C) following a dietary change.
D) following the use of a new probiotic.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

21) More frequent cases of severe *C. difficile* infection since 2003 is attributed to
A) the emergence of more virulent and toxigenic strains.
B) the increased use of probiotics.
C) problems with hospital hygiene and increased transmission.
D) dietary changes in the human population.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

22) Which of the following is TRUE of vaginal bacterial communities?
A) Those dominated by *Lactobacillus crispatus* are more resilient than those dominated by *Lactobacillus iners*.
B) Those dominated by *Lactobacillus crispatus* are less resilient than those dominated by *Lactobacillus iners*.
C) *Lactobacillus iners* is always associated with vaginosis, but *Lactobacillus crispatus* is not.
D) *Lactobacillus crispatus* is always associated with vaginosis, but *Lactobacillus iners* is not.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.9

23) Which of the following is TRUE regarding the interaction between hormones and vaginal microbial diversity?
A) There is a direct relationship between estradiol levels and microbial diversity.
B) There is an inverse relationship between estradiol levels and microbial diversity.
C) There is a direct relationship between luteinizing hormone levels and microbial diversity.
D) There is an inverse relationship between luteinizing hormone levels and microbial diversity.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.9

24) The host-microbiome supraorganism refers to
A) the human body and all of the microbes found within it, but not those on the external surfaces.
B) a nonhuman animal and all of the microbes found within it, but not those on the external surfaces.
C) a human body and all of its commensal microbes.
D) a body and all of its associated microbes.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.1

25) Which of the following is a benefit of studying the human microbiome?
A) It may allow for the development of personalized medical treatments.
B) It may allow for more finely targeted probiotics.
C) It may allow for increased recognition of disease biomarkers.
D) All of these answer choices are possible benefits.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.1

26) Which of the following is TRUE about research on the human microbiome?
A) Advanced nucleic acid sequencing techniques were needed before a good understanding of its microbial abundance and diversity could be developed.
B) Culture-dependent techniques fully elucidated its microbial diversity.
C) Cultivation was unimportant in studying its microbial diversity.
D) None of the answers are correct.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.1

27) Which study set out to generate genomic sequences from 200 strains of bacteria isolated from the human body?
A) the Human Microbiome Project
B) NIH JumpStart Program
C) MetaGenoPolis
D) the Integrative Human Microbiome Project
Answer: B
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.1

28) Based upon microbiome projects, the dominant microbial species on the skin is
A) *Corynebacterium* species.
B) *Streptococcus* species.
C) *Staphylococcus* species.
D) *Propionibacterium* species.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.1

29) Important bacterial populations in the stomach include
A) *Firmicutes*.
B) *Bacteriodetes*.
C) *Veilonella*.
D) All of these answer choices.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

30) The gut microbiota resembles the adult microbiota
A) by 12 months of age.
B) by 1 month of age.
C) around age 15.
D) around age 3.
Answer: D
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.7

31) Normal microbiota helps to ________ colonization of pathogenic organisms.
A) promote
B) prevent
C) maintain
D) accelerate
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8, 24.11

32) The collective term for the functional collection of microbes living on or in the human body, as opposed to a general term for organisms in an environmental habitat, is
A) the human microbiome.
B) the microbiota.
C) transient microbial flora.
D) pathogens.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.1

33) ________ dominate the infant gut microbiota.
A) *Candida*
B) *Actinobacteria*
C) *Firmicutes*
D) *Corynebacteria*
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

34) Which of the following is TRUE about the comparison between the mouse gut and the human gut?
A) The stomach is relatively larger in mice.
B) The colon and cecum are relatively larger in mice.
C) The small intestine is relatively larger in mice.
D) The esophagus and stomach are relatively larger in mice.
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.6

35) The human gastrointestinal tract includes all of the following EXCEPT the
A) small intestine.
B) stomach.
C) large intestine.
D) epiglottis.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

36) Which of the following is known to be TRUE about dog owners and their dogs?
A) Dog owners have fewer skin microbes than those who don't own dogs.
B) Dog owners have more skin microbes than those who don't own dogs.
C) Skin microbes are more common between dog owners and their own dogs than between dog owners and other dogs.
D) Skin microbes are similar between dog owners and dogs in general, with no difference in similarity when dogs and their owners are compared with other human-dog pairs.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.5

37) Infants born via C-section have
A) fewer *Bacteriodes* species than those born vaginally.
B) approximately 72% of the species present matching those of their mothers.
C) relatively less *Staphylococcus aureus* than those born vaginally.
D) fewer *Haemophilus influenzae* than those born vaginally.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

38) Which of the following is the genus represented by the greatest diversity of phylotypes on the skin?
A) *Bacteriodes*
B) *Firmicutes*
C) *Actinobacteria*
D) *Proteobacteria*
Answer: C
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.5

39) Which of the following environmental and host factors influence the composition of the resident microbiota on the skin?
A) age
B) personal hygiene
C) weather
D) age, personal hygiene, and weather
Answer: D
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.5

40) Extensive microbial growth in a thick bacterial layer on the teeth is called
A) dental plaque.
B) dental caries.
C) dental biofilm.
D) periodontitis.
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

41) Gut microbiota is known to be linked with
A) obesity and inflammatory bowel syndrome.
B) obesity, but not with inflammatory bowel syndrome.
C) inflammatory bowel syndrome, but not with obesity.
D) inflammatory bowel syndrome and Crohn's disease only, but not with ulcerative colitis.
Answer: A
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.8

42) What happens when mice grown in the absence of microbes are inoculated with healthy cecal material from other mice?
A) Genes associated with glucose uptake, lipid absorption, and lipid transport are activated and the mice gain body fat without increasing energy intake.
B) Genes associated with glucose uptake, lipid absorption, and lipid transport are activated and the mice gain body fat as a result of taking in more food.
C) Genes associated with protein uptake, lipid break down, and lipid transport are activated and the mice lose body fat without changing their food intake.
D) Genes associated with protein uptake, lipid break down, and lipid transport are deactivated and the mice lose body fat as a result of taking in less food.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.8

43) The Nugent score is used to diagnose
A) urinary tract infections.
B) ulcerative colitis.
C) vaginosis.
D) inflammatory colitis.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

44) Which of the following is NOT true of human gut microbes?
A) The population size is low, but the diversity is high.
B) They produce and excrete amino acids.
C) They help catabolize polysaccharides.
D) They are involved in the "maturing" of the digestive tract.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

45) The human oral microbiota consists of
A) a small group of phylogenetically related aerobic microorganisms.
B) diverse aerobic and anaerobic microorganisms.
C) monoculture biofilms on tooth surfaces.
D) the same phyla that are found in the human gut.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.3

46) Which of the following is NOT a major question in the Human Microbiome Project?
A) Are differences in the relative abundance of different bacteria important?
B) Do differences in the human microbiome correlate with differences in human health?
C) Is there a correlation between microbial population structure and host genotype?
D) How can we reduce the number of microbes on the human body?
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.6

47) The most heavily colonized human organ by bacteria is the ________, containing 10^{11}-10^{12} bacterial cells per gram.
A) mouth
B) small intestine
C) large intestine
D) skin
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

48) Which of the following is NOT a function of the human gut microbiome?
A) production of essential amino acids and vitamins
B) production of volatile fatty acids from polysaccharides
C) iron and trace metal absorption
D) maturation of the gastrointestinal tract
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.3

49) The human microbiome contains all
A) organisms present on the skin.
B) organisms within the digestive system.
C) organisms present in and on the body.
D) organisms present on the skin or in the digestive system only.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.1

50) Weight gain and obesity may be partly caused by certain gut microbial communities that
A) absorb more vitamins and essential amino acids.
B) produce more volatile fatty acids by fermentation.
C) stimulate the gut endothelium to absorb more sugars.
D) produce more hydrogen gas by fermentation.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8

24.2 True/False Questions

1) All microorganisms that live in the human body are harmful.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.1

2) Bacteria found in the mouth in the first years of life are well-adapted to biofilm formation.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.3

3) The presence of *Propionibacterium acnes* is always associated with inflammatory disease.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

4) Normal microbiota is usually found in the blood, lymph, and nervous systems of the body.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.1

5) A vaginal swab would suggest vaginosis if there were relatively high numbers of gram-positive rods and relatively low numbers of gram variable rods.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.9

6) The use of fecal transplants has dramatically improved outcomes for patients with *Clostridium difficile* infections.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

7) *Clostridium difficile* commonly develops in hospitalized patients, but is increasingly seen in non-hospitalized patients traditionally considered to be at lower risk.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

8) Probiotics do not contain living microbes.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.11

9) Protists are NOT normally found in the gastrointestinal tract of healthy individuals.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

10) Bacteria make up about one-third the weight of fecal matter.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

11) *Malassezia* spp. are the most common fungi present on the skin of humans.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.5

12) The upper respiratory tract usually has a considerable amount of resident microflora in a healthy adult.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.3

13) Vaginal acidity in the adult female is due to acid production by *Lactobacillus acidophilus*.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.4

14) Consumers can purchase probiotic foods that are 100% proven to be highly effective.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.11

15) Some yogurts contain probiotics and prebiotics.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.11

16) The male reproductive tract is generally sterile except for the distal portion of the urethra.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.4

17) Goblet cells in the intestine secrete mucin.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

18) Crohn's disease and ulcerative colitis are examples of inflammatory bowel syndromes.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

19) A plant-rich diet increases the risk of inflammatory bowel disease, a leaky gut, and inflammation.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.10

20) Volatile fatty acids are produced primarily from the breakdown of proteins.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8

21) Volatile fatty acids include acetate, proprionate, and butyrate.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8

22) Methanogenic *Archaea* are more common in obese humans compared with thin humans.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8

23) Ammonia-oxidizing Archaea can be found on the skin.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.5

24) Goblet cells in the intestine produce antimicrobial peptides.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2

25) A healthy vaginal microbiome is dominated by *Candida*.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.4

26) Elderly individuals have greater proportions of *Bacteroidetes* in their gut microbiota compared with younger individuals.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

27) The gut microbiota is relatively stable over long time periods in healthy adults, suggesting its highly evolved role in the human body.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

28) Biofilms in the human mouth primarily contain pathogens.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.3

24.3   Essay Questions

1) Explain how diet plays a role in host susceptibility to infection, focusing on the role of diet in the development of inflammatory bowel disease.
Answer: Diet alters the normal flora, allowing opportunistic pathogens a better chance of multiplying and increasing the overall susceptibility of the host to known pathogens. For example, inflammatory bowel disease (IBD) is strongly correlated with diet and is caused by an imbalance in the normal microbiota. For example, a diet rich in animal protein can contribute to the presence of harmful metabolites in the large intestine that contribute to leaky gut and inflammation.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.8

2) Why are some microorganisms specialized to only certain parts of the body? Describe an example that supports your answer.
Answer: Each organ differs chemically and physically from others and thus provides a selective environment for the growth of certain microorganisms. Examples will vary.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.1

3) Explain how the composition of breast milk compared with formula affects the microbiota of infants.
Answer: Breast milk contains oligosaccharides that encourage the growth of *Bifobacterium longum*. Additionally, pathogens bind these oligosaccharides rather than to similar carbohydrates on the wall of the infant gastrointestinal tract. As a result, pathogens are less likely to become established. Formula-fed infants tend to have higher populations of *Clostridium difficile, Enterobacter cloacae, Citrobacter* spp., *Granulicatella adiacens*, and *Bilophila wadsworthia*.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.7

4) Explain how prebiotics and probiotics may be useful in preventing acne.
Answer: By using prebiotics, it is possible to add nutrients that may encourage the growth of nonpathogens. Probiotics add bacteria to the skin that may take up residence and reduce the risk of colonization by pathogenic strains. Both prebiotics and probiotics may help to alter the skin microbiota enough to prevent the growth of strains associated with acne around the growing colonies.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.7, 24.11

5) Describe how oral antibiotics can result in the harmful alterations in the microbiota.
Answer: Antibiotics inhibit the growth of the normal flora as well as pathogens, leading to the loss of antibiotic-susceptible bacteria in the intestinal tract. In the absence of the full complement of normal flora, opportunistic microorganisms can become established.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.8, 24.9

6) What is the similarity between the microbiota found in a female at the menopause stage and a female at the pre-puberty stage? How different is it from a female between the stages of puberty and menopause?
Answer: The microbiota in both cases lack *Lactobacillus acidophilus*. This species ferments glygocen, causing a slightly acidic pH. Before puberty and after menopause, glycogen is not present and the pH is more neutral.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 24.4

7) Compare the normal microbiota in the male and female urogenital tracts.
Answer: The male urogenital tract is normally sterile except for the distal part of the urethra. The female urinary tract is also usually sterile except for the distal part of the urethra, but the vagina has a complex microbiota that varies along with hormonal and other changes.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.4

8) Discuss some reasons why it is important to include diverse populations in studies of the human microbiome.
Answer: The microbiome is highly variable and is influenced by both genetics and environment. Because it is so variable, understanding it requires studying people from a variety of genetic and geographical backgrounds. It is important to consider people of different ethnicities and with different health conditions, such as obesity. Only by considering a diverse range of participants will it be possible to deeply understand the normal microbiota and how it is affected by environmental and genetic factors.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 24.6

9) What bacteria are important in the human gut and what evidence do we have that specific bacteria present are important for human health?
Answer: Important phyla in the human gut are *Firmicutes*, *Bacteriodetes*, *Proteobacteria*, and *Actinobacteria*. Evidence exists that there may be three different types of gut communities, or enterotypes, in the human population. These enterotypes seem to have different metabolic pathways that affect the processing of nutrients and the health of the gut. Gut microbes synthesize nutrients (e.g., amino acids and vitamins) that are important in the human diet. One example of a taxon affecting human health is the suggestion that the presence of *Prevotella* (a genus of *Bacteriodetes*) may contribute to obesity through increased production of volatile fatty acids that can be used as a source of energy (calories). Also, an increase in the number of methanogens in the gut may increase fermentation and the production of VFAs, thus increasing the number of calories available. (Other examples would be acceptable here as well.)
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 24.2